#ManyDaughters Project: Pre-Analysis Plan

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1 H1: Having daughters leads to stronger support for gender equality (as measured by attitudes).

1.1 Variables

1.1.1 Dependent variable:

After a detailed review of the available variables across survey waves, the item **plh0395i05**: "*Schools and preschools should promote non-gender-typical activities that are not typical for children's gender*", emerges as a strong proxy for attitudes toward gender equality in the general population.
While other items, such as **plh0395i04**: "*Sex assignment is not so important for a person's development*", could also serve as indicators of such attitudes, I consider **plh0395i05** to be preferred due to its clear alignment with the core objective of challenging gender stereotypes and promoting an environment in which individuals can realize their potential free from traditional gender constraints.[1] This rejection of normative assumptions based on gender, actually reflects a foundational principle of gender equality. Moreover, this item captures a proactive stance toward reshaping socialization processes and represents a progressive, preventive approach to addressing the root causes of gender inequality.
The variable is measured on a 7-point Likert scale ranging from "[1] Do not agree at all" to "[7] Completely agree". A significant limitation is that **plh0395i05** is included only in the 2019 wave, which precludes the use of the panel dimension of the SOEP and restricts the analysis to a cross-sectional perspective. Despite this constraint, the item yields 25,017 valid responses with substantial variation in the distribution of answers, ranging from 5.2% for the least selected category to 34.4% for the most frequent one, indicating sufficient heterogeneity for meaningful analysis.

1.1.2 Independent variable of interest:

A binary variable, **has_daughters**, will be created, coded as 1 if the respondent has at least one *biological* daughter born before the interview (2019, **imonth**), and 0 otherwise. To construct this variable, fertility history information from the SOEP biographical instrument will be used. In particular, SOEP provides information on both the year (**kidgeb**k) and the month (**kidmon**k) in which child $k \in \{1, \ldots, 19\}$ was born and the child's sex (**kidsex**k). Thus,

- **has_daughters**$_i = 1$ if for respondent i there exists at least one $k \in \{1, \ldots, 19\}$ such that **kidgeb**$k < 2019$ and **kidsex**$k = 2$.
- **has_daughters**$_i = 1$ if the above doesn't hold but there exists $k \in \{1, \ldots, 19\}$ such that **kidgeb**$k = 2019$ and **kidsex**$k = 2$ and **kidmon**$k \leq$**imonth**.

[1] In addition, the alternative item **plh0395i04** is less suitable due to its conceptual ambiguity. The statement "Sex assignment is not so important for a person's development" is unclear in terms of whether it is intended to be interpreted descriptively (i.e., as a statement about how society currently functions) or normatively (i.e., as a claim about how things should be). This ambiguity may lead to inconsistent responses: a respondent committed to gender equality might strongly agree from a normative standpoint, believing that sex assignment should not constrain individual development, while the same respondent might strongly disagree if interpreting the item descriptively, recognizing that, in practice, sex assignment still plays a significant role in shaping life outcomes. This interpretive uncertainty undermines its reliability as an indicator of attitudes toward gender equality.

- **has_daughters**$_i = 0$ otherwise.

Notice that this variable will only capture the extensive but not the intensive margin, which is enough to test the hypothesis.
The decision to include only biological daughters stems from the fact that the effect of having daughters on respondent's attitudes toward gender equality may be determined importantly by the existence of a direct and potentially formative relationship which sit is plausible that is more direct and potent in the case of a biological daughter, as it involves a more fundamental and often lifelong experience of parenthood. In contrast, stepdaughters may enter the respondent's life at a later stage, potentially limiting the extent of shared experiences and reducing the impact on the respondent's attitudes. Adoption, on the other hand, may introduce selection bias: families with stronger gender-equal attitudes could be more likely to adopt girls, confounding the relationship of interest. Finally, according to the SOEP documentation, it appears that determining the existence of biological daughters is simple and reliable in the data, while determining the existence of non-biological daughters is more complicated.

1.1.3 Control Variables:

- $Kids_i$: This variable will take value 1 if the number of children reported to have been born before the interview is at least 1 and 0 otherwise. In particular, this variable will be constructed by assigning it a value of 1 if the date of birth of the first child (**kidgeb01**, **kidmon01**) is prior to the interview date (2019, **imonth**).

- $Male_i$: Recovered from the variable **sex**, this will be recorded as a binary variable (0 = Female, 1 = Male).

- Age_i: Calculated as 2019−**gebjahr**. Used as a continuous variable.

- $Education_i$: Information recovered from variable **pgisced11** which classifies all correspondents' educational degrees according to the "International Standard Classification of Education (ISCED)" of 2011. It would take into account the information for year 2019. Used as a factor variable.

- $Marital_Status_i$: Recovered from the **pgfamstd** variable. It would take into account the information at year 2019. Used as factor variable.

- $Origin_i$: Country in which subject i was born. Recovered from the variable **corigin**. Used as fixed effect.

- $Contacted_i$: First year that subject i was contacted to answer the survey. Recovered from the variable **eintritt**. Since different waves are run for specific profiles, it seems interesting to control for wave-fixed effects.

1.2 Exclusion criteria

- Only data with birth biography will be used. Thus, only those with **biovalid**=30 or **biovalid**=31 will be used for the analysis.

- Observations with missing values for any of the above listed variables will be excluded.

1.3 Analysis:

Although the dependent variable is an ordinal scale ranging from 1 to 7, an OLS regression will be performed. Standard errors would be clustered at the **hid** (Current Wave Household Number) level, since it is possible that attitudes toward gender equality are correlated within the household itself due to unobservable factors and/or common influences.
The analysis to be carried out is

$$plh0395i05_i = \beta_0 + \beta_1 \cdot has_daughters_i + \theta \cdot X_i + \epsilon_i$$

where X_i is a respondent i's vector of the control variables as described above while θ is the corresponding vector of coefficients associated with them.

1.3.1 Interpretation of results:

The coefficient β_1 associated to the variable *has_daughters* is expected to be positive and statistically significant. A positive coefficient would indicate that having daughters is associated with greater support for gender equality, in contrast to having sons (The effect of having sons is captured by the coefficient on the variable *Kids*).

2 H2: Having daughters leads to more left-leaning political preferences.

2.1 Variables:

2.1.1 Dependent variable:

The item **plh0004** — "*Political attitude: In politics people often talk about "left" and "right" when it comes to characterizing different political attitudes. If you think about your own political views: Where would you place yours?*" — appears to be a suitable measure to summarize left-leaning political preferences. This item is measured on a scale from [0] to [10], where [0] indicates a far-left position and [10] indicates a far-right position. Accordingly, the variable will be reverse-coded so that higher scores represent more left-leaning political preferences. In particular, the variable $\textbf{plh0004_R}_{i,t}$ will be computed as $\textbf{plh0004_R}_{i,t} = 10 - \textbf{plh0004}_{i,t}$.
This item was included in the 2005, 2009, 2014, and 2019 survey waves, yielding a total of 90,972 valid responses and exhibiting substantial variation between subjects.

2.1.2 Independent variable of interest:

A binary variable, **has_daughters** will be defined for each respondent i at each period $t \in \{2005, 2009, 2014, 2019\}$ similar to how it is defined in Hypothesis 1. In particular:

- $\textbf{has_daughters}_{i,t} = 1$ if for respondent i there exists at least one $k \in \{1, \ldots, 19\}$ such that $\textbf{kidgeb}k < t$ and $\textbf{kidsex}k = 2$.
- $\textbf{has_daughters}_{i,t} = 1$ if the above doesn't hold but there exists $k \in \{1, \ldots, 19\}$ such that $\textbf{kidgeb}k = t$ and $\textbf{kidsex}k = 2$ and $\textbf{kidmon}k \leq \textbf{imonth}_t$.
- $\textbf{has_daughters}_{i,t} = 0$ otherwise.

Same caveats to the ones discussed in Hypothesis 1 apply.

2.1.3 Control Variables:

- $Kids_{i,t}$: This variable will take value 1 if the number of children reported by respondent i to have been born before the interview performed on year $t \in \{2005, 2009, 2014, 2019\}$ is at least 1 and 0 otherwise. In particular, this variable will be constructed by assigning it a value of 1 if the date of birth of the first child (**kidgeb01**, **kidmon01**) is prior to the realization of the survey in year t (t, **imonth**).
- $Age_{i,t}$: Calculated as $t - \textbf{gebjahr}$. Used as a continuous variable.
- $Education_{i,t}$: Information recovered from variable **pgisced11** which classifies all correspondents' educational degrees according to the "International Standard Classification of Education (ISCED)" of 2011. It would take into account the information for year t. Used as a factor variable.
- $Marital_Status_{i,t}$: Recovered from the **pgfamstd** variable at year t. It would take into account the information at year t. Used as factor variable.

2.2 Inclusion/exclusion criteria:

- Only data with birth biography will be used. Thus, only those with **biovalid**=30 or **biovalid**=31 will be used for the analysis
- Since the dependent variable is only available for years 2005, 2009, 2014 and 2019, only data available for this year will be used.
- Observations with missing values for any of the above listed variables will be excluded.

2.3 Analysis:

To test this hypothesis, the dataset will be structured as a longitudinal panel, with multiple observations for the same individuals across the survey waves in which the dependent variable is available (2005, 2009, 2014, and 2019). A fixed-effects regression model will be employed to control for unobserved, time-invariant individual characteristics that may be correlated with both having daughters and political preferences, thereby mitigating potential endogeneity concerns related to stable individual traits:

$$plh0004_R_{i,t} = \beta_1 \cdot has_daughters_{i,t} + \theta \cdot X_{i,t} + \gamma_t + \alpha_i + \epsilon_{i,t}$$

where $X_{i,t}$ is a vector of control variables for individual i at time t, γ_t denotes year fixed effects (for 2005, 2009, 2014, and 2019), α_i captures unobserved, time-invariant individual fixed effects, and $\epsilon_{i,t}$ is the idiosyncratic error term in year t. Standard errors will be clustered at the **hid** level (Current Wave Household Number).

2.4 Interpretation of results:

The coefficient β_1 associated with the variable *has_daughters* is expected to be positive and statistically significant. A positive estimate of β_1 would suggest that having daughters is associated with more left-leaning political preferences, relative to having sons. The effect of having sons is controlled for through the inclusion of the variable *Kids* in the vector X_{it} of control variables.

3 H3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

3.1 Variables:

3.1.1 Dependent variable:

In the PL dataset, there are two key variables capturing the time individuals spend on unpaid household tasks:[2]

- **pli0043_h**: Number of hours the respondent spends on *housework* (laundry, cooking, cleaning) on a typical weekday.
- **pli0044_h**: Number of hours the respondent spends on *childcare* on a typical weekday.

In a couple composed of subject i and subject j, the contribution of subject i to household chores in household h can be computed as:

$$\frac{pli0043_h_i + pli0044_h_i}{(pli0043_h_i + pli0044_h_i) + (pli0043_h_j + pli0044_h_j)}$$

Gender equality in the division of household labor would imply that, within couples and on average, men and women contribute equally to these tasks, i.e., the above indicator equals 0.5. Any systematic deviation from 0.5 would reflect gender inequality in the allocation of unpaid household work.
Therefore, a measure of gender equality in household h can be computed as:

$$Equality_h = -\left|0.5 - \frac{pli0043_h_i + pli0044_h_i}{(pli0043_h_i + pli0044_h_i) + (pli0043_h_j + pli0044_h_j)}\right|$$

where $Equality_h \in [-0.5, 0]$, with -0.5 indicating the highest degree of inequality (one individual performs all the work), and 0 indicating full equality. Couples will be identified using the variable **pgpartnr**.
Since the variables **pli0043_h**, **pli0044_h**, and **pgpartnr** are available throughout the entire period from 2000 to 2020, the indicator $Equality_h$ can be constructed for this full time span.

3.1.2 Independent variable of interest:

A binary variable, **has_daughters**, will be defined for each respondent i in each period $t \in \{2000, \ldots, 2020\}$, similarly to the definition used in Hypothesis 1. Specifically:

[2]Other variables in the dataset also capture unpaid work, but I chose to focus on **pli0043_h** and **pli0044_h** because these two tasks (housework and childcare) can, in principle, be performed interchangeably by men and women. Admittedly, this may not fully apply at certain stages of child-rearing (e.g., breastfeeding), but in such cases, a greater share of the workload can be compensated by the partner through increased housework, making equity achievable if desired. In contrast, other tasks included in the questionnaire, such as "*Errands*" (**pli0040**), are more likely to be performed on a personal basis and are not easily interchangeable (e.g., handling bureaucratic procedures). Moreover, these tasks may not always serve the collective interest of the household, but rather benefit a single member (e.g., shopping for individual needs).

- **has_daughters**$_{i,t} = 1$ if, for respondent i, there exists at least one $k \in \{1, \ldots, 19\}$ such that **kidgeb**$k < t$ and **kidsex**$k = 2$.
- **has_daughters**$_{i,t} = 1$ if the above condition does not hold but there exists $k \in \{1, \ldots, 19\}$ such that **kidgeb**$k = t$, **kidsex**$k = 2$, and **kidmon**$k \leq$ **imonth**$_t$.
- **has_daughters**$_{i,t} = 0$ otherwise.

The same caveats discussed in Hypothesis 1 apply here.

A household-level variable, **has_daughters**$_{h,t}$, indicating whether the household h (composed of respondents i and j) is exposed to having at least one daughter, will be defined as:

$$\textbf{has_daughters}_{h,t} = \begin{cases} 1 & \text{if } \textbf{has_daughters}_{i,t} = 1 \text{ or } \textbf{has_daughters}_{j,t} = 1 \\ 0 & \text{otherwise} \end{cases}$$

3.1.3 Control variables:

Control variables are grouped into two categories: those related to individual characteristics of the household members and those pertaining to the household itself.

- **Individual-level controls** for the woman ($X^w_{h,t}$) and the man ($X^m_{h,t}$) in household h:
  - *Kids*$_{i,t}$: A binary variable equal to 1 if respondent i has at least one child born before the date of the interview in year t, and 0 otherwise. In particular, it is set to 1 if the date of birth of the first child (**kidgeb01**, **kidmon01**) precedes the survey month (**imonth**) in year t.
  - *Age*$_{i,t}$: Calculated as $t-$ **gebjahr**. Treated as a continuous variable.
  - *Education*$_{i,t}$: Coded using the **pgisced11** variable, which follows the 2011 International Standard Classification of Education (ISCED). Used as a categorical (factor) variable.
  - *Marital_Status*$_{i,t}$: Derived from **pgfamstd** at year t. Treated as a categorical variable.
  - *Employment_Status*$_{i,t}$: Derived from **pgemplst** at year t. Treated as a categorical variable.
- **Household-level controls** for household h:
  - *Income*$_{h,t}$: Retrieved from variable **hghinc** in year t, reporting the monthly net household income.
  - *Residence*$_{h,t}$: Based on variable **hgnuts1**, indicating the NUTS1 region of residence.

3.2 Inclusion/exclusion criteria:

- Only individuals with **biovalid**$= 30$ or **biovalid**$= 31$ will be included in the analysis.
- The analysis will be restricted to heterosexual couples. This restriction is necessary for the definition of the dependent variable and the specific hypothesis focusing on male-female dynamics within a household.

- Only households in which a couple cohabits will be considered. To ensure this, observations with **pgpartnr**$= -2$ will be excluded, as this code is assigned to individuals without a spouse or (cohabiting) partner (equivalent to codes 0, 3, 4, 5 in the variable **pgpartz**).

3.3 Exact analysis:

To test this hypothesis, the dataset will be structured as a longitudinal panel at the household level, with repeated observations for the same household across different survey waves (from 2000 to 2020). A *household* is defined as a unique and stable pairing of individuals i and j. That is, for a given household, the composition in terms of individuals i and j remains constant over time, although any given individual may belong to different households across waves (never more than one in a given wave).
For consistency and interpretability, household h is defined as comprising a woman i and a man j. While the gender composition does not affect the symmetry of the equality index, this convention allows us to consistently control for potential gender differences in the effects of personal characteristics (e.g., education) on $Equality_h$.
In line with this data structure, a fixed-effects regression model will be employed to account for unobserved time-invariant household characteristics that may be correlated with gender equality.

$$Equality_{h,t} = \beta_1 \cdot \text{has_daughters}_{h,t} + \theta \cdot X^w_{h,t} + \phi \cdot X^m_{h,t} + \psi \cdot X_{h,t} + \gamma_t + \alpha_h + \epsilon_{h,t}$$

where $X^w_{h,t}$ is the vector of control variables for the female component of household h at time t, $X^m_{h,t}$ is the vector of control variables for the male component of household h at time t, $X_{h,t}$ is the vector of control variables for household h at time t, γ_t are year fixed effects for survey years 2000 to 2020, α_h is the unobserved, time-invariant household fixed effect, and $\epsilon_{h,t}$ is the idiosyncratic error term for household h in year t. Standard errors will be clustered at the **hid** (Current Wave HH Number) level.[3]

3.4 Interpretation of results:

Since the $Equality_h$ variable is constructed such that higher values (closer to 0) indicate greater gender equality within the household, a positive and statistically significant coefficient ($\beta_1 > 0$) on the $has_daughters_h$ variable would provide evidence in support of the hypothesis.

[3]The variable **hid** will be used for clustering standard errors provided it remains consistent over time, as suggested by the documentation available at https://companion.soep.de/. However, if this interpretation is incorrect and household identifiers are in fact year-specific even when the household composition remains unchanged, then an alternative identifier will be constructed by combining the personal identifiers (**pid**) of individuals i and j.

4 H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

4.1 Variables

Same as in Hypothesis 1.

4.2 Exclusion criteria

Same as in Hypothesis 1.

4.3 Analysis

As in Hypothesis 1, an OLS regression will be estimated with standard errors clustered at the **hid** level. The key difference is that the respondent's gender will be considered as a moderating variable. Specifically, the hypothesis will be tested using the following specification:

$$plh0395i05_i = \beta_0 + \beta_1 \cdot has_daughters_i + \beta_2 \cdot male_i + \beta_3 \cdot (has_daughters_i \cdot male_i) + \theta \cdot X_i + \epsilon_i$$

where X_i denotes the vector of control variables for respondent i (as described in Hypothesis 1), and θ is the corresponding vector of coefficients.

4.3.1 Interpretation of results:

The coefficient β_3 on the interaction term $has_daughters_i \cdot male_i$ is expected to be positive and statistically significant. This would indicate that the effect of having daughters ($has_daughters_i = 1$) on support for gender equality ($plh0395i05_i$) is stronger for men ($male_i = 1$) than for women.